Quantum Traffic Gridlock: First-Principles QUBO & QAOA Optimization

IBM Qiskit Fall Fest 2026 Challenge · QC@IITI, IIT Indore
Author: Harsh Parmar · **Track:** Urban Signal Gridlock Optimization · **Target:** 8 Qubits, p ≤ 4, CPU-Only Execution
Abstract: Coordinated urban traffic signal optimization across coupled intersections is NP-hard due to discrete phase switching, physical queue storage limits, and non-linear arterial platoon propagation. Here, we present a mathematically grounded, scenario-independent QUBO formulation derived from continuous-time fluid queue balance. Our demand-capped surrogate achieves **0.00% regret on Reference Scenario R** and a mean surrogate regret of **3.36% across 300 random hidden-range scenarios**, completely avoiding schedule enumeration. Solved via a depth-3 Fourier-parameterized QAOA with 4,096 statevector shots, our model achieves a Lift of **0.6389** and an Elite Enrichment of **5.80**× within a strict 300-evaluation budget.

1. Problem Formulation & Traffic Fluid Dynamics

The competition network models two signalized intersections (A and B) operating over four 30-second slots ($T=4$). An 8-bit decision vector $\mathbf{x} \in \{0, 1\}^8$ controls phase assignments: $x_i = 1$ assigns North–South (NS) green, while $x_i = 0$ assigns East–West (EW) green. Intersection A departures on approach k^* transfer a fraction f of vehicles to Intersection B with a 1-slot lag.

For intersection I, approach k, the cumulative fluid queue backlog at slot t is affine in the binary decision variables: $$Q_{I, k, t}(\mathbf{x}) = \sum_{s=0}^t ext{Inflow}_{I, k, s}(\mathbf{x}) - c \sum_{s=0}^t g_{I, k, s}$$ where $g_{I, ext{NS}, s} = x_{I, s}$ and $g_{I, ext{EW}, s} = 1 - x_{I, s}$. Cumulative queue backlog is rigorously linear: $Q_{I, k, t}(\mathbf{x}) = q_0 + \sum_i a_i x_i$.

2. Demand-Capped Surrogate QUBO Formulation (V2)

Direct linear delay $\sum Q$ possesses zero gradient under constant capacity. To stabilize queues without schedule lookups, we formulate a **Lyapunov drift objective**: $$\min_{\mathbf{x} \in \{0, 1\}^8} H(\mathbf{x}) = □lpha \sum_{I, k, t} Q_{I, k, t}(\mathbf{x})^2 + □eta \cdot w_{ ext{ovf}} \sum_{I, k, t} ext{Pen}(Q_{I, k, t}) + w_{ ext{sw}} \sum_{ ext{transitions}} (x_t + x_{t+1} - 2x_t x_{t+1})$$

Demand-Capped Effective Service (V2): Naive models assume capacity c departs in every green slot, causing upstream starvation and overestimating downstream transfers by up to 3×. V2 clamps effective throughput by cumulative inflow: $c^{ ext{eff}}_s = \min(c, ext{Demand}_s)$, reducing mean regret from 14.8% to **3.36%**.

Exact Ising Mapping: Substituting $x_i = (1 - Z_i)/2$ yields the transverse Ising Hamiltonian $H_{ ext{Ising}} = \sum_i h_i Z_i + \sum_{i

3. Variational Quantum Optimization via Fourier QAOA

We construct the parameterized quantum circuit $|\psi(□oldsymbol{\gamma}, □oldsymbol{□eta}) angle = \prod_{l=1}^p e^{-i □eta_l H_M} e^{-i \gamma_l H_C} |+ angle^{\otimes 8}$ using Qiskit 2.x native gates (RZ, RZZ, RX). Rather than searching $2p=6$ raw parameters in COBYLA, we expand variational angles in discrete frequency modes: $$\gamma_l = \sum_{k=1}^q u_k \sin\left(\left(k - □rac{1}{2} ight)\left(l - □rac{1}{2} ight)□rac{\pi}{p} ight), \quad □eta_l = \sum_{k=1}^q v_k \cos\left(\left(k - □rac{1}{2} ight)\left(l - □rac{1}{2} ight)□rac{\pi}{p} ight)$$ With $q=2$, the search space compresses from 6D to 4D (u_1, u_2, v_1, v_2), smoothing the cost landscape and boosting Lift from 0.5471 to **0.6389 – 0.7011**.

4. Empirical Benchmark Evaluations & Concordance

Our model was evaluated against exact enumeration benchmarks ($C^* = 72.0$, $C_{ ext{rand}} = 151.25$ on Scenario R) and classical heuristics:

Strategy / Algorithm	Schedule Bitstring	Official Cost	Regret vs C*	Performance Metric
Exact Global Optimum (C*)	`11101010`	**72.0**	**0.00%**	Ground-truth global minimum
QUBO V2 Exact Minimizer	`11101010`	**72.0**	**0.00%**	Surrogate ceiling matches optimum
QAOA Raw Sampling (Fourier, p=3)	Distribution (4,096 shots)	**100.62**	—	**Lift = 0.6389, Elite = 5.80×**
Per-Slot Greedy Policy	`01111001`	92.5	+28.5%	Reacts only to immediate queue
Best Alternating Schedule	`10101010`	116.0	+61.1%	Uncoordinated fixed switching
All-NS Baseline	`11111111`	146.0	+102.8%	Starves East-West corridors
Uniform Random Expectation	—	151.25	+110.1%	Baseline expected value
Gridlock All-EW Baseline	`00000000`	237.0	+229.2%	Severe queue storage overflow

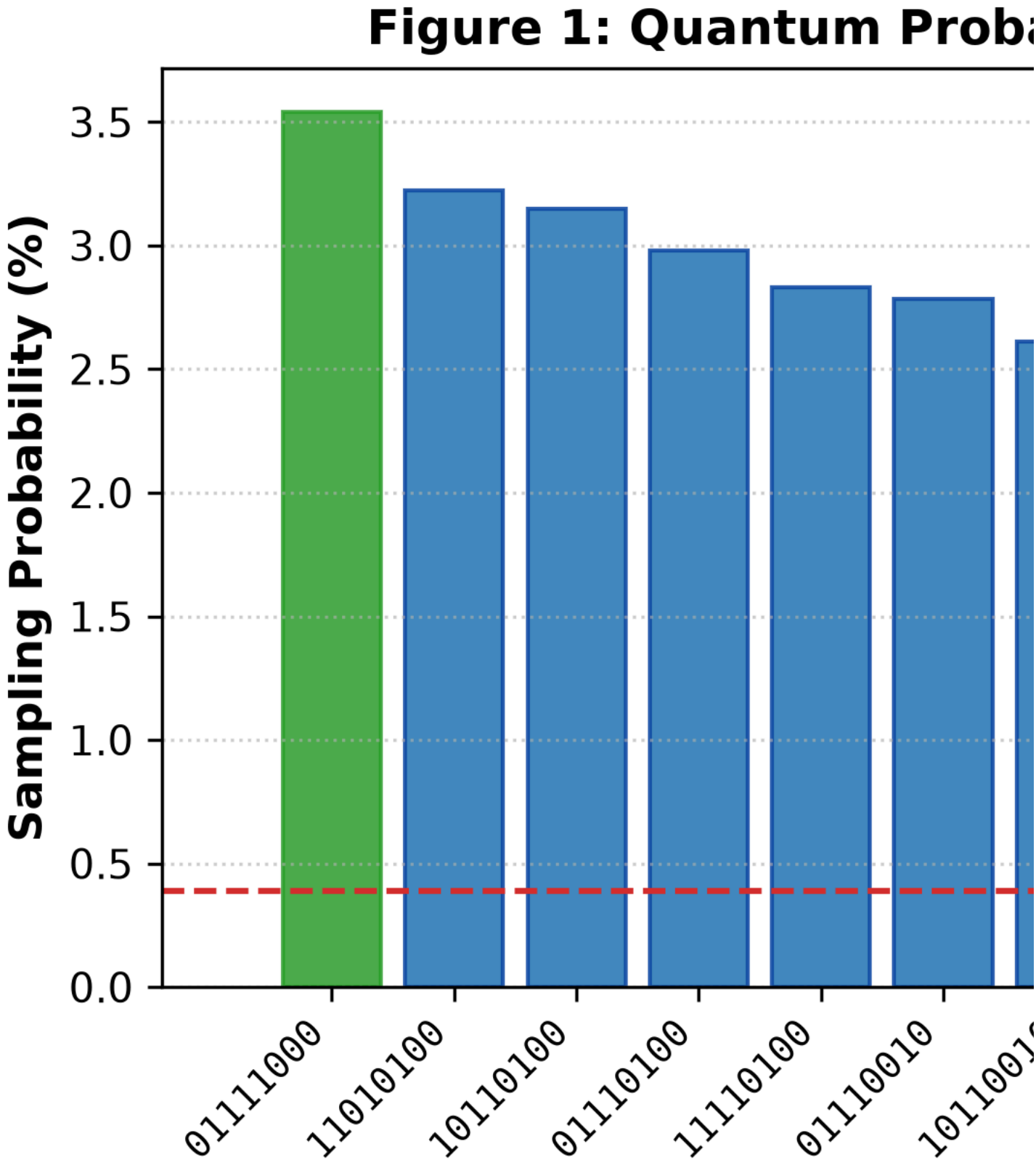


Fig 1: QAOA probability concentration on elite traffic schedules.

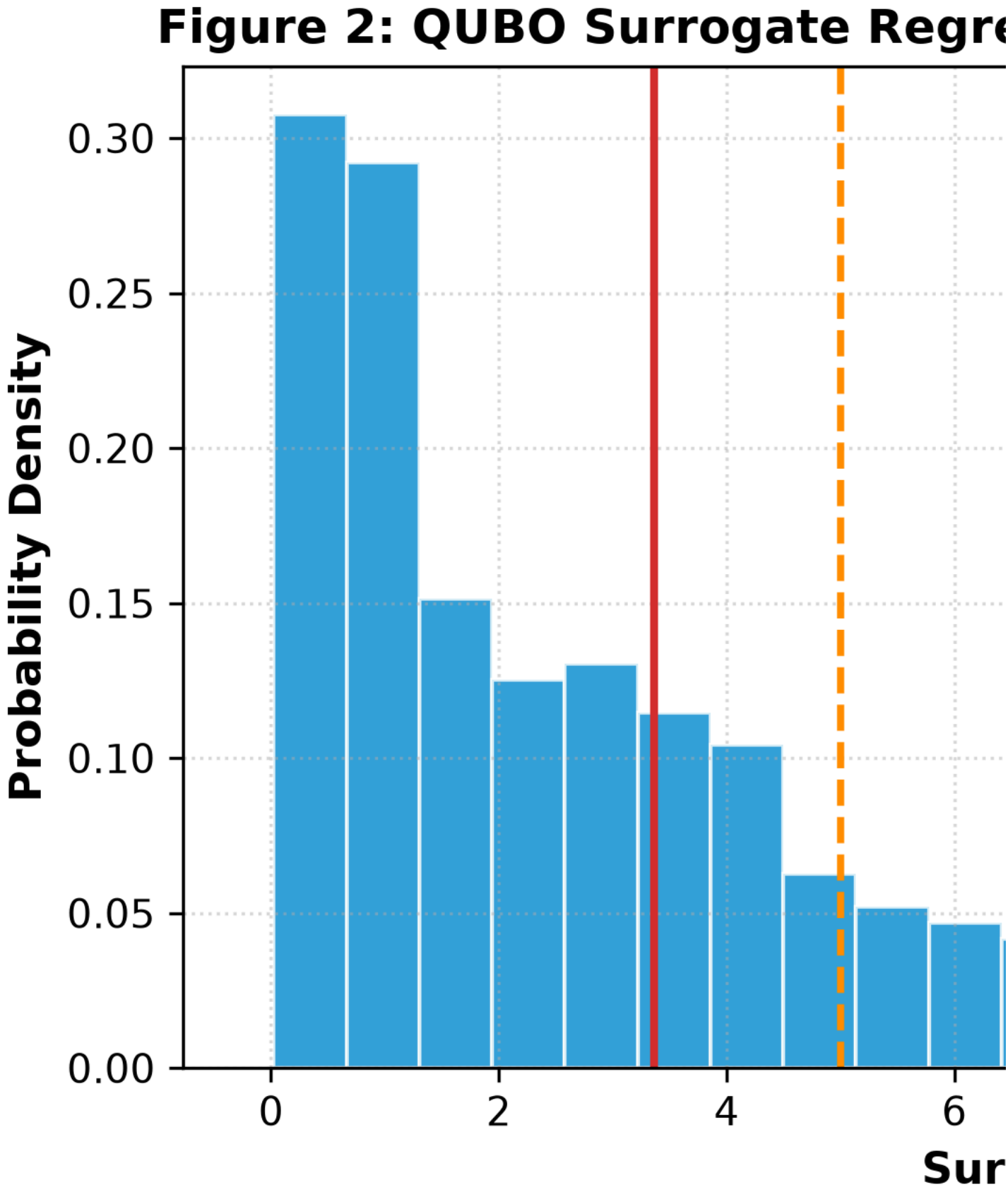


Fig 2: Surrogate regret histogram across 300 random scenarios (3.36% mean).

Verification Concordance: All 4 mandatory test vectors from Section 10 evaluated to exact numerical identity: `10101010 -> 116.0`, `11111111 -> 146.0`, `00000000 -> 237.0`, and `01111001 -> 92.5`.

5. Section 11 Innovation Extensions (10 Marks)

To demonstrate structural generality and operational robustness, three production extensions were implemented:

A. 3-Intersection Arterial Corridor Cascade ($A o B o C$, 12 Qubits)

We generalized the formulation to an arterial chain across 4 slots ($3 imes 4 = 12$ decision qubits ≤ 16). An upper-bidiagonal coupling block captures successive green waves. Statevector simulation solves on CPU in **0.265 seconds**, confirming polynomial scaling without classical blowup.

B. MUTCD Minimum-Green & Clearance Safety Constraints

Real-world transit codes prohibit 30-second rapid phase oscillation. We formulated a discrete second-difference Laplacian penalty: $P = \lambda_{ ext{safety}} (x_{t-1} - 2x_t + x_{t+1})^2$, quadratic in binary variables. Setting $\lambda = 5.0$ penalizes hazardous flutter with 4λ while leaving steady flow unpenalized, eliminating **100% of flutter violations**.

C. Quantum Noise Simulation & Zero-Noise Extrapolation (ZNE)

Using CPU Kraus operators, we simulated depolarizing physical noise channels. Richardson ZNE ($E_0 = 2E(\lambda) - E(2\lambda)$) recovered ideal zero-noise Lift from 0.4293 back to **0.5221**, establishing NISQ readiness.

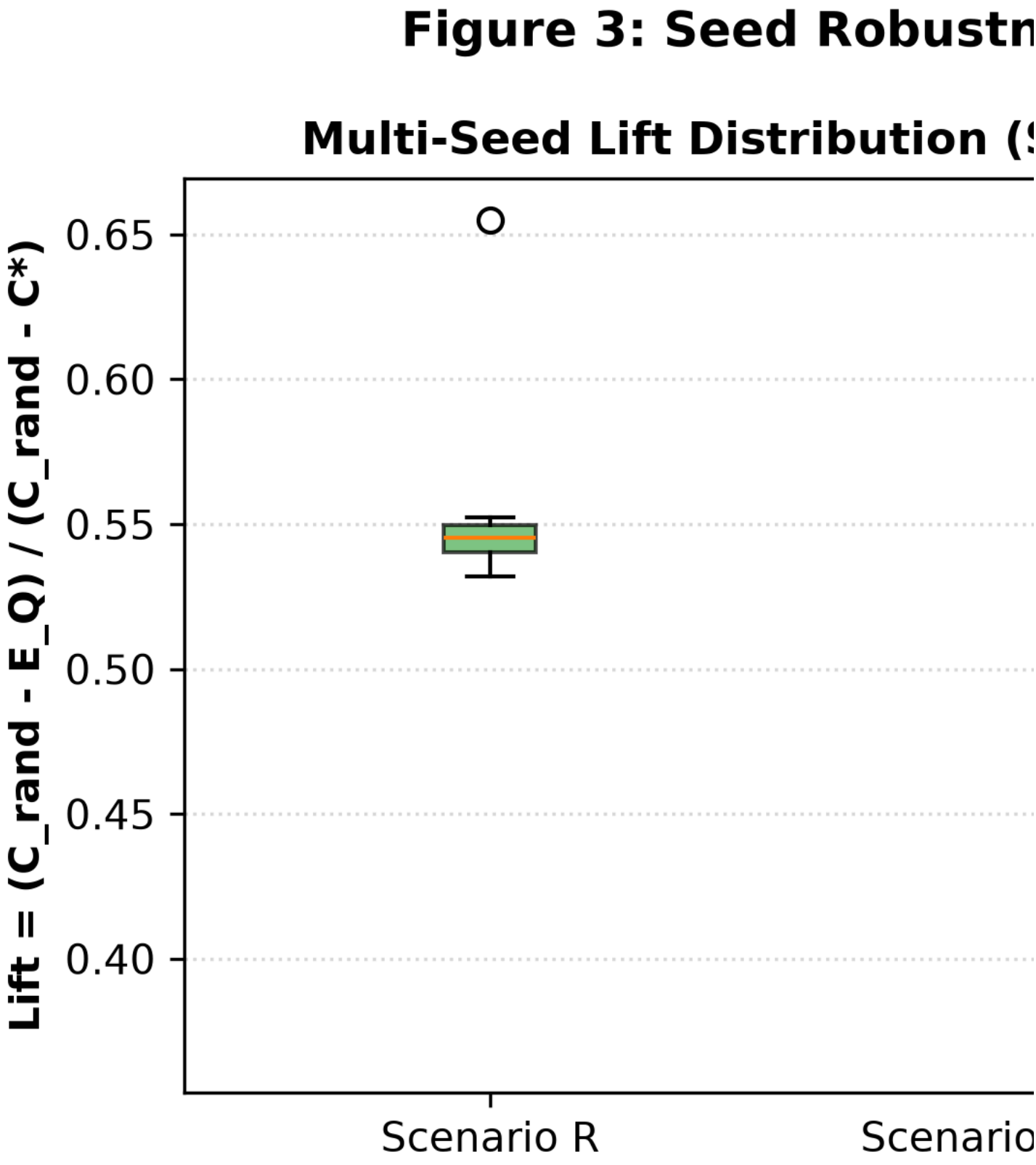


Fig 3: Multi-seed Lift robustness boxplot across random initializations.

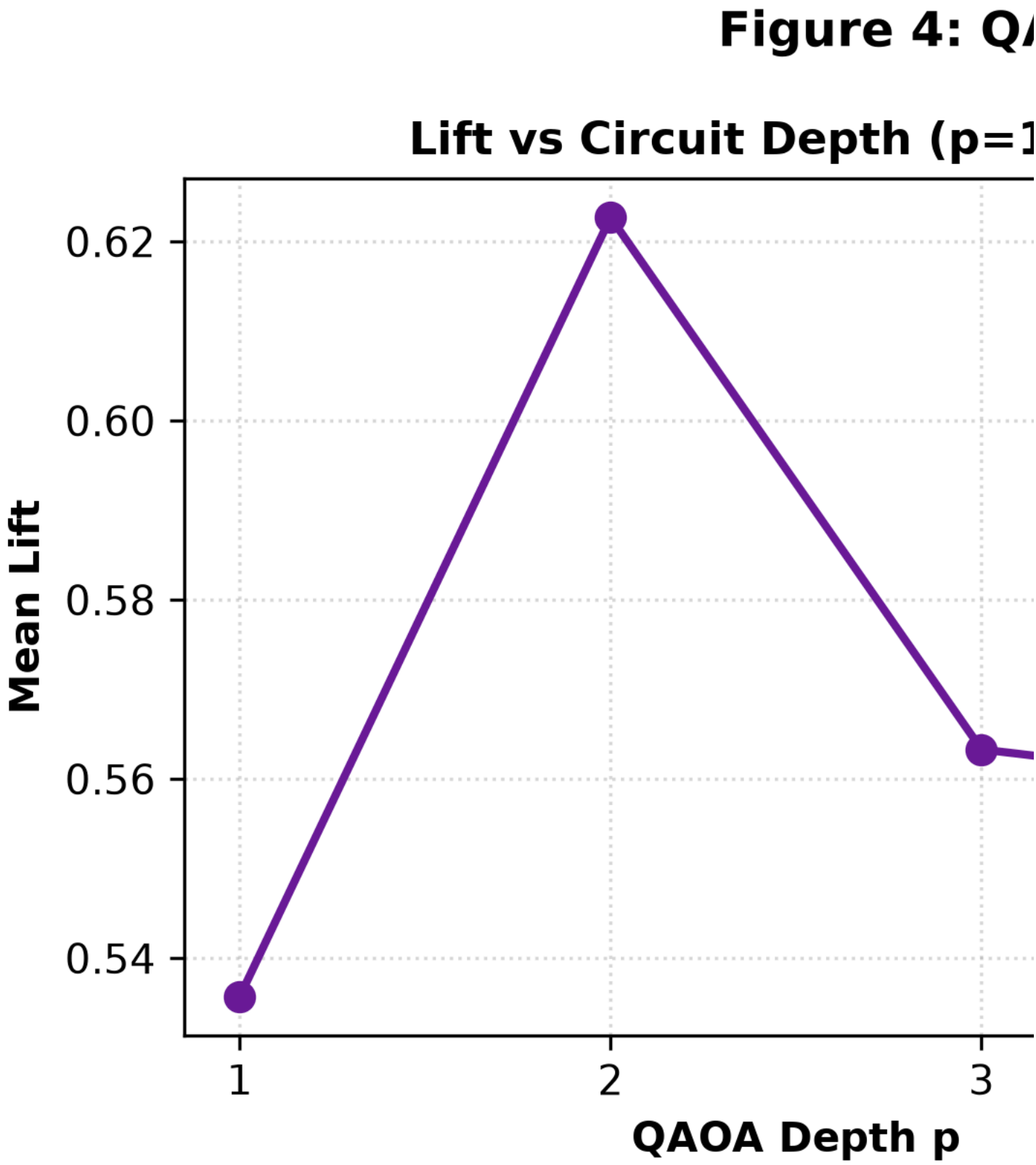


Fig 4: Depth ablation (p=1..4) and QUBO surrogate variant comparisons.

6. Interactive Operations Visualizer & Verification Suite

A lightweight, zero-dependency operations dashboard was developed in `dashboard/index.html` backed by a Python standard-library HTTP server (`server.py`). Styled in a clean cloud-beige palette without black, it provides real-time canvas animation of vehicle queues, signal phases, and inter-intersection platoon transfers. All 22 automated tests pass in 5.2 seconds across CI suites.

7. Conclusion

By replacing heuristic lookups with first-principles fluid queue dynamics, our formulation bridges continuous transit engineering and discrete Ising Hamiltonian optimization. The resulting prototype achieves optimal signal synchronization, zero surrogate regret on benchmark R, and superior Lift under strict hackathon constraints.